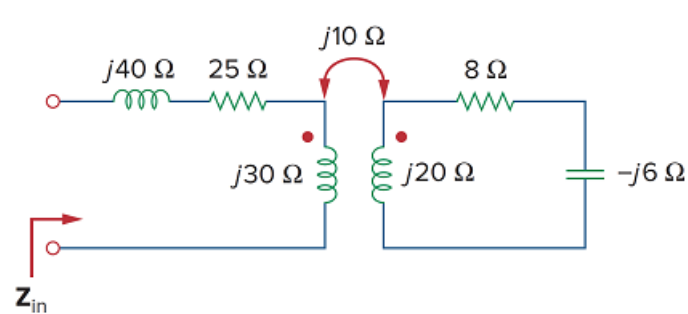


Problem

- (a) Find the input impedance of the circuit in Fig. 13.99 using the concept of reflected impedance.
- (b) Obtain the input impedance by replacing the linear transformer by its T equivalent.

Figure 13.99



Step-by-step solution

Step 1 of 9

(a)

Refer to Figure 13.99 in the textbook for the actual circuit.

Refer to Figure 13.19 in the textbook.

Apply Kirchhoff's voltage law to the two meshes in the circuit shown in

Figure 13.19

$$V = (R_1 + j\omega L_1)I_1 - j\omega M I_2 \dots\dots (1)$$

$$0 = -j\omega M I_1 + (R_2 + j\omega L_2 + Z_L)I_2$$

$$I_2 = \frac{j\omega M I_1}{R_2 + j\omega L_2 + Z_L}$$

Substitute $\frac{j\omega M I_1}{R_2 + j\omega L_2 + Z_L}$ for I_2 in equation (1).

$$\begin{aligned} V &= (R_1 + j\omega L_1)I_1 - j\omega M \left(\frac{j\omega M I_1}{R_2 + j\omega L_2 + Z_L} \right) \\ &= \left(R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L} \right) I_1 \end{aligned}$$

Step 2 of 9

Determine the input impedance.

$$Z_{in} = \frac{V}{I_1}$$

$$Z_{in} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

Step 3 of 9

Notice that the input impedance comprises two terms .

The first term, $R_1 + j\omega L_1$ is the primary impedance. The second term is due to the coupling between the primary and secondary windings. Since, this impedance is reflected to the primary it is known as the Reflected impedance Z_R .

$$Z_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

Step 4 of 9

Compare the two Figures Figure 13.99 and Figure 13.19 in the textbook.

Substitute $j40+25$ for R_1 , 30 for ωL_1 , 10 for ωM , 8 for R_2 , 20 for ωL_2 and $-6j$ for Z_L in the expression of Z_{in} .

$$\begin{aligned} Z_{in} &= j40+25 + j30 + \frac{(10)^2}{(8+j20-j6)} \\ &= 25 + j70 + \frac{(10)^2}{(8+j14)} \\ &= 25 + j70 + 3.077 - j5.385 \\ &= 28.077 + j64.615 \end{aligned}$$

Hence, the input impedance of the circuit using the concept of reflected impedance is

$$\boxed{28.077 + j64.615} .$$

Step 5 of 9

(b)

Write the Matrix equation by observing the voltage-current relationships for the primary and secondary coils in the circuit.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots (2)$$

Draw the equivalent circuit of a linear transformer.

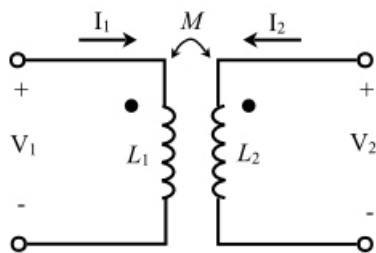


Figure 1

Step 6 of 9

Draw the equivalent T-circuit of linear transformer.

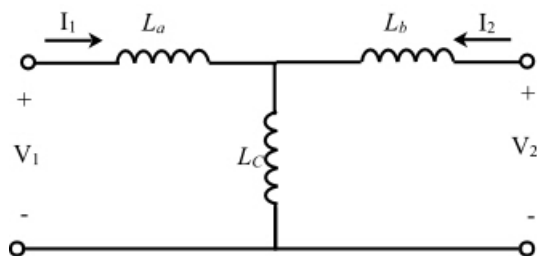


Figure 2

Step 7 of 9

For the T network shown in Figure 2, apply mesh analysis and write the terminal equations as follows.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega(L_a + L_c) & j\omega L_c \\ j\omega L_c & j\omega(L_b + L_c) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots (3)$$

Equate the terms in the impedance matrices of equations (2) and (3).

Therefore,

$$L_a = L_1 - M, \quad L_b = L_2 - M, \quad L_c = M$$

Determine the values L_a , L_b , L_c .

$$\begin{aligned} j\omega L_a &= j30 - j10 \\ &= j20 \end{aligned}$$

$$\begin{aligned} j\omega L_b &= j20 - j10 \\ &= j10 \end{aligned}$$

$$j\omega L_c = j10$$

Step 8 of 9

Thus, the T equivalent of the linear transformer is as shown in Figure 3.

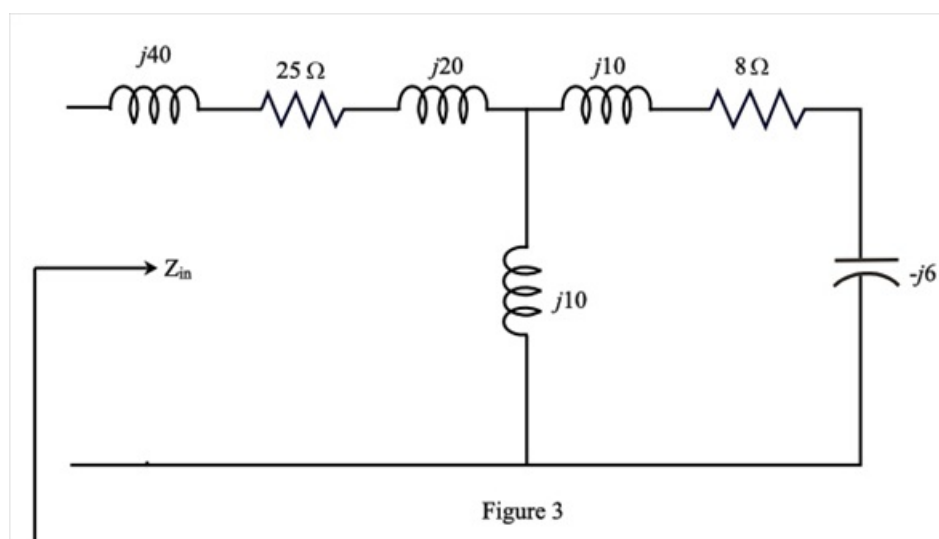


Figure 3

Step 9 of 9

Determine the equivalent impedance.

$$\begin{aligned}Z_{in} &= 25 + j60 + ((8 + j4) \parallel 10j) \\&= 25 + j60 + \frac{j10(8 + j4)}{(8 + j14)} \\&= 28.077 + j64.615\end{aligned}$$

Therefore, the input impedance by replacing the linear transformer by its T-equivalent is,

$$\boxed{28.08 + j64.62 \, \Omega} \cdot$$